

1943

Aqua-Lung

Frenchmen Jacques Cousteau and Emile Gagnan invented the Aqua-Lung, a self-regulating underwater breathing apparatus. Their portable invention popularized diving. It featured a regulator that adjusted air pressure and managed air supply so that the pressure of air inside a diver's lungs matched with that of the surrounding water.

The diver breathes in air through a tube from the cylinder.

Cylinder contains air

Jacques Cousteau (right) with another diver, wearing Aqua-Lungs



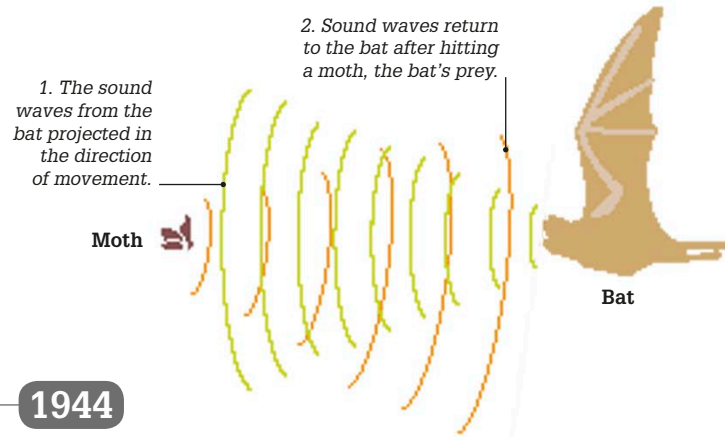
Work on Colossus—a pioneering early electronic computer used to break German codes and secret messages—began in 1943.

1. The sound waves from the bat projected in the direction of movement.

Moth

2. Sound waves return to the bat after hitting a moth, the bat's prey.

Bat



1944

Echolocation discovered

American biophysicist Donald Griffin coined the term echolocation to describe how bats (as well as some whales, dolphins, and shrews) emit sound waves to navigate and hunt prey. The sound reflects back off objects and is analyzed by the animal's brain fast and accurately, helping it detect prey. Some species of bat can catch more than 500 insects per hour using echolocation.

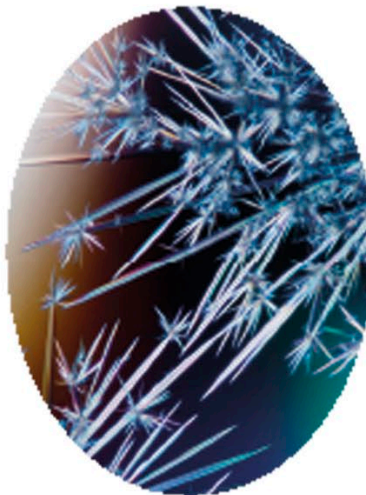
1945



1943

Antibiotic tackles tuberculosis

The chemical streptomycin was first isolated from bacteria found in soil by staff in the laboratory of American scientist Selman Waksman. It proved to be the first antibiotic to successfully combat the disease tuberculosis. Streptomycin has since been used to treat other diseases such as tularemia, and as a pesticide against certain fungal diseases of fruit crops.



Streptomycin crystals under a microscope

1943

Kidney dialysis

Dutch physician Willem Johan Kolff built the first kidney dialysis machine using materials such as orange juice cans and an old washing machine. The machine worked as an artificial kidney, taking the blood out of a patient whose own kidneys were not working, filtering the blood to remove harmful toxins, and then returning clean blood to the patient.

1945

Microwave oven invented

While working on RADAR (Radio Detection And Ranging) technology at a company called Raytheon, American physicist Percy Spencer discovered how high-powered vacuum tubes called magnetrons gave off microwaves (a type of electromagnetic wave). These waves cause the molecules in food to vibrate and heat up. Spencer and Raytheon developed the first commercial microwave oven, the Radarange.



Hamburger cooked in an original Raytheon Radarange microwave oven